

IT3708 - Bio-Inspired Artificial Intelligence

Ole J. Mengshoel
ole.j.mengshoel@ntnu.no

Professor, Department of CS, NTNU, Trondheim

Adjunct Faculty, Department of ECE, CMU, Silicon Valley

May 5th, Spring 2021 Semester

Examples of Exam Questions with Answers

Question 1

☐ 1. Multiple Choice: Among the following options, which op...

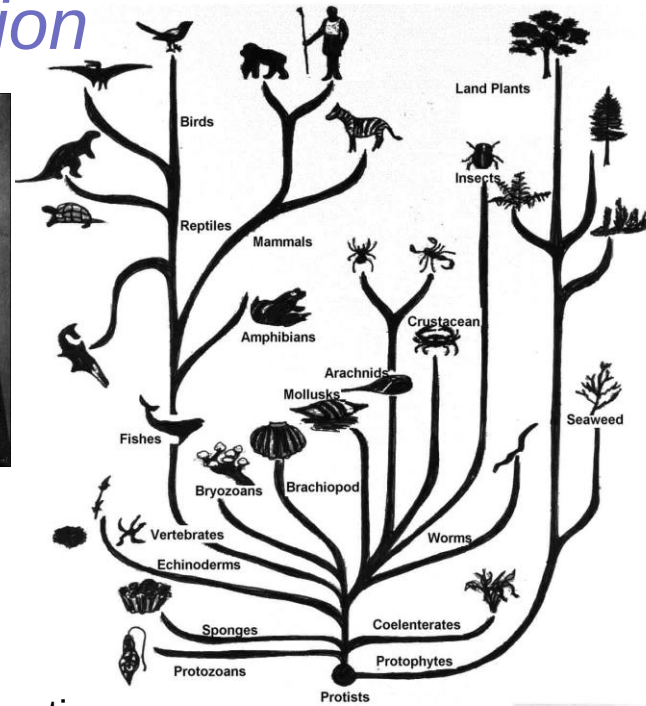
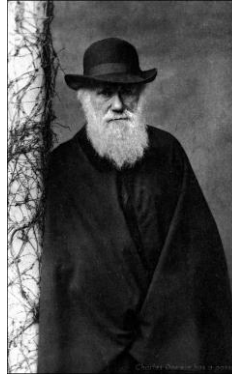
Points: 1

Question	Among the following options, which option is not considered a "pillar of evolution"?
Answer	Population Diversity Heredity  Constraints Selection

The 4 Pillars of Evolution

All species derive from
common ancestor

Charles Darwin, 1859
On the Origins of Species



Population

Group of several individuals

Diversity

Individuals have different characteristics

Heredity

Characteristics are transmitted over generations

Selection

- Individuals make more offspring than the environment can support
- Better at food gathering = better at surviving = make more offspring

Question 2

☐ 2. Multiple Choice: Among the following operations, which...

Points: 1

Question	Among the following operations, which is not part of an evolutionary algorithm's generation cycle?
Answer	Mutation Evaluation  Initialization Crossover Selection Reproduction

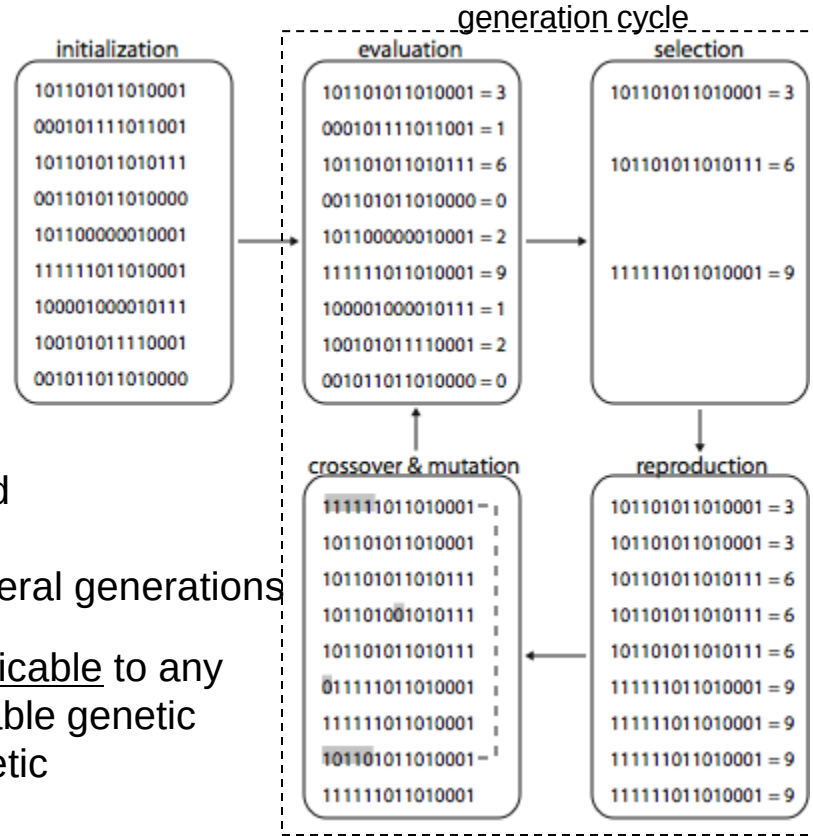
Evolutionary Algorithm

- Devise genetic representation
- Build a population
- Design a fitness function
- Choose selection method
- Choose crossover & mutation
- Choose data analysis method

Repeat generation cycle until:

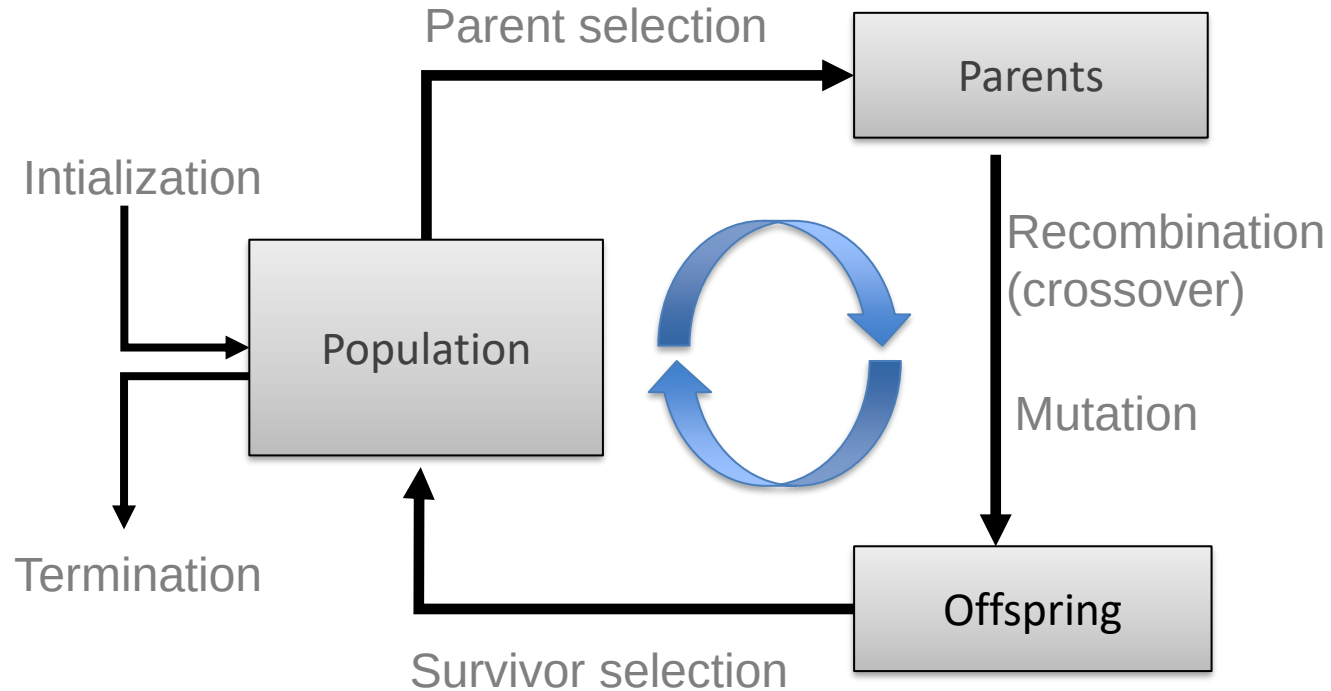
- maximum fitness value is found
- solution found is good enough
- no fitness improvement for several generations

Evolutionary algorithms are applicable to any problem domain as long as suitable genetic representation, fitness, and genetic operators are chosen.



Scheme of an EA:

General scheme of EAs



Question 3

☐ 3. Multiple Choice: In which type of genetic algorithm (G...

Points: 1

Question	In which type of genetic algorithm (GA) is one individual generated per generation, replacing one individual in the population?
Answer	Crowding GA Generational GA  Steady-state GA Messy GA Elitist GA

Population Management Models: Introduction

- Two different population management models exist:
 - *Generational genetic algorithm* (GGA)
 - each individual survives for exactly one generation
 - the entire set of parents is replaced by the offspring
 - *Steady-state genetic algorithm* (SSGA)
 - one offspring is generated per generation
 - one member of population replaced
- Most EAs use either the GGA or the SSGA mode
 - The simple genetic algorithm (SGA) of Goldberg uses GGA
 - Whitley's *GENITOR* is a steady-state GA - SSGA
- Generation Gap (GG)
 - The proportion of the population replaced at each iteration
- How parameter GG varies:
 - For GGA: $GG = 1$
 - For SSGA: $GG = 1/\text{pop_size}$

Question 4

□ 4. Multiple Choice: What is a potential unfortunate conse...

Points: 1

Question	What is a potential unfortunate consequence of too high selection pressure in an evolutionary algorithm that is performing optimization?
Answer	 The population converges to individuals at or close to a local optimum that is not a global optimum. The fitness of the population does not increase much. All individuals in the population converge to be exactly alike, at a local optimum that is not a global optimum. All individuals in the population converge to be exactly alike, at a global optimum. The population converges to individuals at or close to a global optimum.

Selection and Selection Pressure: Summary

- Intuition: Selection plays a role in creating selection pressure in the Evolutionary Algorithm
- *Selection pressure*:
 - Can be defined mathematically as [Simon 2013, p. 199] as:
$$\Phi = \text{Pr}(\text{selection of best fit}) / \text{Pr}(\text{selection of average fit})$$
 - But can also be understood more intuitively
- *Too high selection pressure*: The problem of premature convergence, the population converges to individuals at a local optimum
- *Too low selection pressure*: The problem of too little convergence, the fitness of the population doesn't increase much – optima are not found before the algorithms terminates
- [Simon 2013, p. 199]: «If the selection method is overly biased towards selecting fit individuals, then the population may converge too quickly to a uniform solution [...]. However, if a selection methods is not biased strongly enough towards fit individuals, then the EA may not be able to properly exploit the information that is present in the most fit individuals»

Question 5

☐ 5. Multiple Choice: Which of the following is not considered...

Points: 1

Question	Which of the following is not considered a space in which to preserve diversity in an evolutionary algorithm?
Answer	Algorithm space ✔ Probability space Genotype space Phenotype space

Approaches for Preserving Diversity: Introduction (1/2)

Different spaces for preserving diversity:

- Genotype space
 - Set of representable solutions
- Phenotype space
 - The end result
 - Neighbourhood structure may bear little relation with genotype space
- Algorithmic space
 - Equivalent of the geographical space on which life on earth has evolved
 - Structuring the population of candidate solutions

Question 6

☐ 6. Multiple Choice: Which of the following is not considered...

Points: 1

Question	Which of the following is not considered a diversity-preserving genetic algorithm?
Answer	Deterministic crowding genetic algorithm Fitness sharing genetic algorithm Probabilistic crowding genetic algorithm  Simple genetic algorithm Island model parallel genetic algorithm

Crowding in GAs

- Crowding applies in the survival stage of GAs; phases:
 - *Grouping*: individuals are grouped (paired), often but not necessarily with their offspring, according to a similarity metric
 - *Replacement (competition)*: for each group (pair), one or more winners are picked – remain in the population
- Crowding variants in evolutionary algorithms:
 - Preselection [Cavicchio, 70]
 - Crowding factor [DeJong, 75]
 - Simulated annealing [Kirkpatrick et al., 83]
 - Gene-invariant GA [Culberson, 92]
 - Deterministic crowding [Mahfoud, 95]
 - Parallel recombinative SA [Mahfoud & Goldberg, 95]
 - Restricted Tournament Selection [Harik, 95]
 - Probabilistic crowding [Mengshoel & Goldberg, 99]
 - Generalized crowding [Galan & Mengshoel, 10]
 - Adaptive generalized crowding [Mengshoel et al., 13]
 - ...

Explicit Approaches for Preserving Diversity:

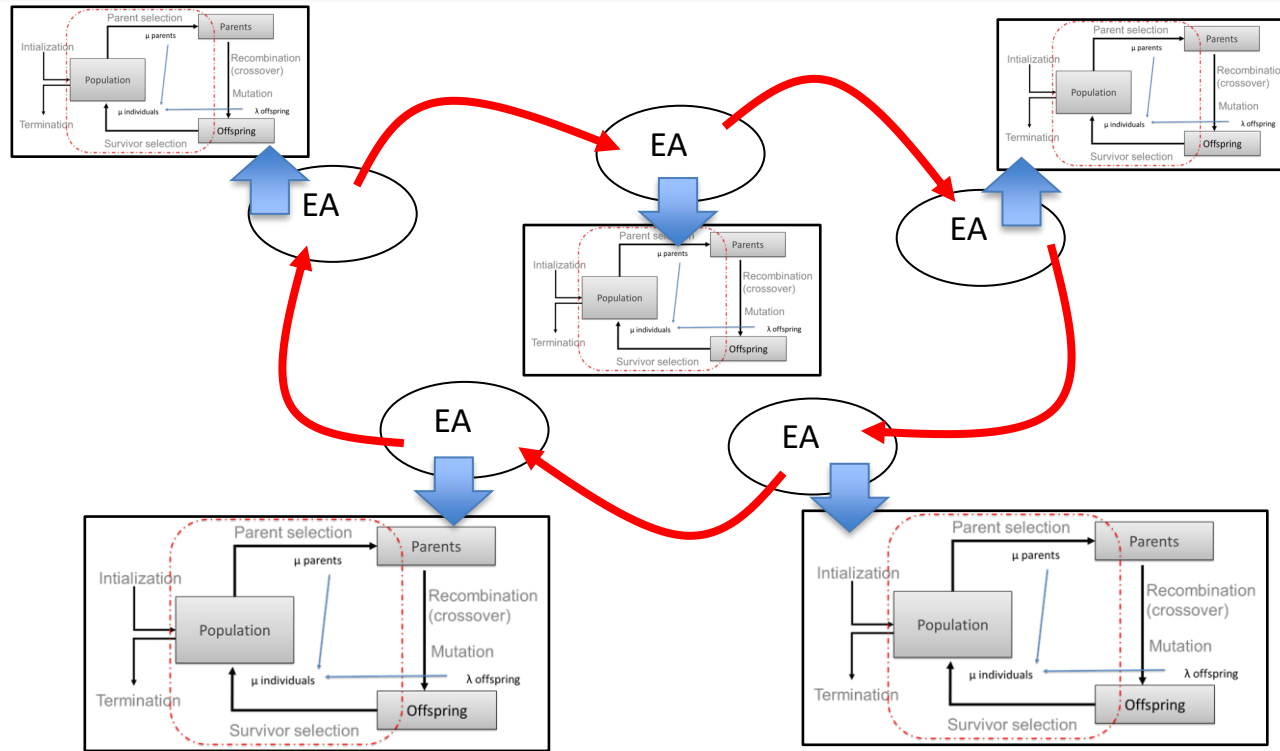
Fitness Sharing (1/2)

- Restricts the number of individuals within a given niche by “sharing” their fitness:
 - Niche fitness is a shared resource – think sheep eating grass
 - Individuals are allocated to a niche **in proportion to the niche fitness**
- Need to set the size of the niche σ_{share} in either genotype or phenotype space
 - Here we assume genotype space
- Run EA as normal but after each generation set

$$f'(i) = \frac{f(i)}{\sum_{j=1}^{\mu} sh(d(i, j))}$$

$$sh(d) = \begin{cases} 1 - (d / \sigma_{\text{share}})^{\alpha} & d \leq \sigma_{\text{share}} \\ 0 & \text{otherwise} \end{cases}$$

Implicit Approaches for Preserving Diversity: “Island” Model Parallel EAs (1/3)



→ : Periodic migration of individual solutions between populations

Question 7

7. Multiple Choice: Local search or stochastic local sear...

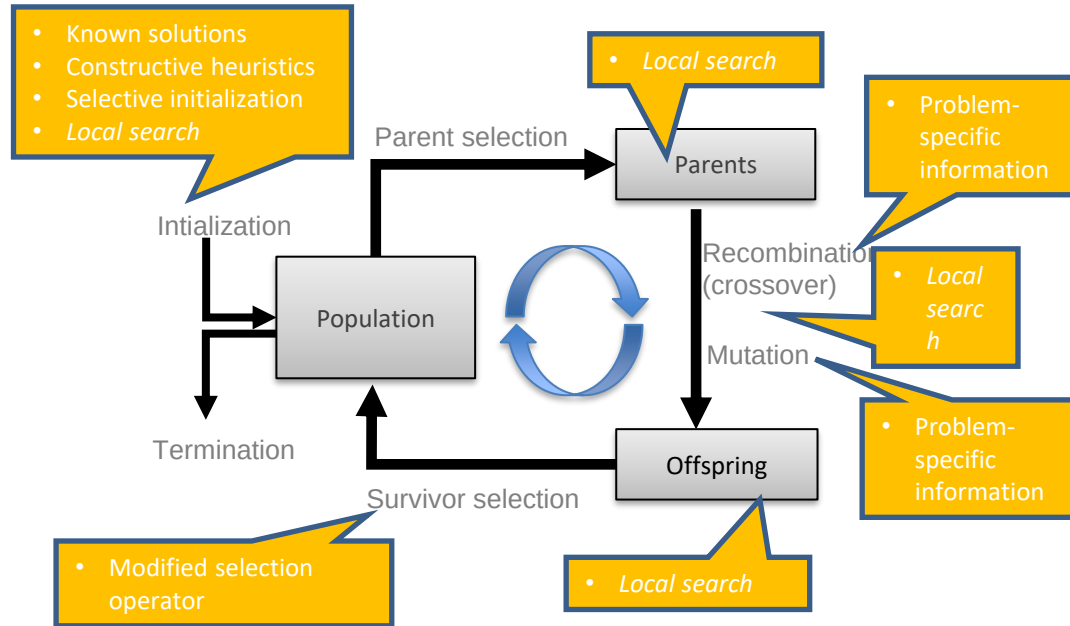
Points: 2

Question	Local search or stochastic local search methods are often used for hybridising evolutionary algorithms (EAs). In which of the following EA components can, in principle, local search or stochastic local search methods be integrated?
Answer	For initialization only. For initialization or survivor selection only.  For initialization, after parent selection, after mutation, or after crossover. For initialization or after parent selection, but never after mutation. For initialization or after parent selection, but never after crossover.

Incorporating good solutions: Heuristics for Initialising Population

- Bramlette ran experiments with limited time scale and suggested holding a n -way tournament amongst randomly created solutions to pick initial population
(n.b. NOT the same as taking the best $popsiz$ of $n.popsiz$ random points)
- **(Stochastic) Local Search (SLS) is another option: pick $popsiz$ points at random to climb from**
 - We have discussed SLS already
 - Another example with Bayesian networks: Later
- Constructive Heuristics often exist, for example:
 - “Nearest neighbour” for TSP
 - “Schedule hardest first” for scheduling and planning

Where and How to Hybridise



Question 8

8. Multiple Choice: Parameter setting is important in evo...

Points: 1

Question	Parameter setting is important in evolutionary algorithms, and there are different methods for setting parameters. Consider the following description of parameter setting: "a feedback rule based on a measure that monitors search progress." Which type of parameter setting method is this?
Answer	Parameter tuning Deterministic parameter control  Adaptive parameter control Self-adaptive parameter control

Question 9

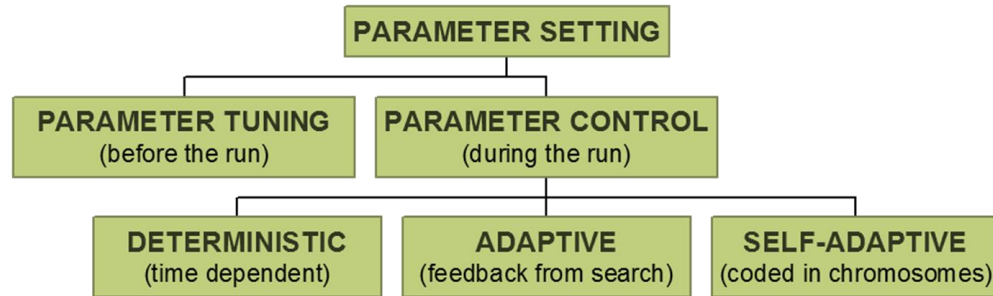
9. Multiple Choice: Parameter setting is important in evo...

Points: 1

Question	Parameter setting is important in evolutionary algorithms, and there are different methods for setting parameters. Consider the following description of parameter setting: "parameter values evolve along with individuals; the values are encoded in the chromosomes and therefore they undergo variation and selection along with the rest of the chromosome." Which type of parameter setting method is this?
Answer	Parameter tuning Deterministic parameter control Adaptive parameter control  Self-adaptive parameter control

How to apply parameter control

Global taxonomy



How to apply parameter control

Three major types of parameter control:

- **deterministic**: some rule modifies strategy parameter without feedback from the search (based on some counter)
- **adaptive**: feedback rule based on some measure monitoring search progress
- **self-adaptive**: parameter values evolve along with solutions; encoded onto chromosomes they undergo variation and selection

Question 10

10. Multiple Choice: Crowding methods are well-established...

Points: 2

Question	Crowding methods are well-established methods for preserving diversity in evolutionary algorithms (EAs). Which of the following statements about crowding is not true?
Answer	The competition phase of an evolutionary algorithm is where crowding differs the most from traditional EAs. In generalized crowding, the trade-off between exploration and exploitation can be varied by using a scaling factor parameter. When the scaling parameter is 0, generalized crowding equals deterministic crowding. ✔ When the scaling parameter is 2, generalized crowding equals probabilistic crowding. Adaptive generalized crowding employs a method for parameter control.

Generalized Crowding

- Parameter ϕ – a scaling factor – is introduced
- Probability P_c of **child** c winning over **parent** p :

$$P_c = \begin{cases} \frac{f(c)}{f(c) + \phi \times f(p)} & \text{if } f(c) > f(p) \\ 0.5 & \text{if } f(c) = f(p) \\ \frac{\phi \times f(c)}{\phi \times f(c) + f(p)} & \text{if } f(c) < f(p) \end{cases} \quad (\text{Eq. 1})$$

where $\phi \in \mathbb{R}^+ \cup \{0\}$ is the scaling factor

- Idea: fitness scaling of the least fit among **p (parent)** and **c (child)** is done before “Probabilistic Crowding” is applied

The Idea of Generalized Crowding

- We consider competition phase of crowding
- **The trade-off between exploration and exploitation can be varied using a parameter ϕ - scaling factor**
 - ϕ is non-negative, real-valued
 - high exploitation, low exploration: $\phi = 0$
 - low exploitation, high exploration: high ϕ
- The “Generalized” part of Generalized Crowding:
 - $\phi = 0$: Generalized Crowding = Deterministic Crowding
 - $\phi = 1$: **Generalized Crowding = Probabilistic Crowding**
- Importantly, $\phi = 0$ or $\phi = 1$ are not always best
 - key motivation for **Adaptive** Generalized Crowding

Question 11

11. Multiple Choice: This problem is concerned with the mi...

Points: 1

Question

This problem is concerned with the minimization of two fitness functions f_1 and f_2 .

In particular, consider the points $\{x_1, x_2, x_3, x_4\}$ and their fitness function values for a multiobjective minimization problem:

- $f_1(x_1) = 1, f_2(x_1) = 1,$
- $f_1(x_2) = 1, f_2(x_2) = 2,$
- $f_1(x_3) = 2, f_2(x_3) = 1,$ and
- $f_1(x_4) = 2, f_2(x_4) = 2.$

Which point dominates all the others?

Answer

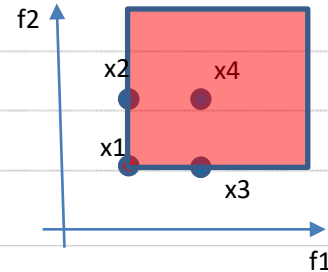
✓ x1

x2

x3

x4

None



Question 12

12. Multiple Choice: This problem is concerned with the mi...

Points: 1

Question

This problem is concerned with the minimization of two fitness functions f_1 and f_2 .

In particular, consider the points $\{x_1, x_2, x_3, x_4\}$ and their fitness function values for a multiobjective minimization problem:

- $f_1(x_1) = 1, f_2(x_1) = 1,$
- $f_1(x_2) = 1, f_2(x_2) = 2,$
- $f_1(x_3) = 2, f_2(x_3) = 1,$ and
- $f_1(x_4) = 2, f_2(x_4) = 2.$

Which point is nondominated?

Answer

✓ x1

x2

x3

x4

None

